

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2021 – Paper 7

7 Further Graphics (aco41)

The Rendering
Equation

- (a) Recall that the (local) rendering equation is given as $L_o(\mathbf{x}, \vec{\omega}_o) = L_e(\mathbf{x}, \vec{\omega}_o) + \int_{H^2} f_r(\mathbf{x}, \vec{\omega}_i, \vec{\omega}_o) L_i(\mathbf{x}, \vec{\omega}_i) \cos \theta_i d\vec{\omega}_i$. Simplify this expression progressively as much as possible after each of the following assumptions. [5 marks]
- (i) There is no light emitted from this surface point $\mathbf{x}$.
 - (ii) Diffuse reflection.
 - (iii) Reflection is constant at all surface points.
 - (iv) The incoming light is the same for all incident angles but there can be occluders.
 - (v) There is a single object and we ignore self-occlusions.

Answer:

- (i) There is no light emitted from this surface point $\mathbf{x}$.

$$L_o(\mathbf{x}, \vec{\omega}_o) = \int_{H^2} f_r(\mathbf{x}, \vec{\omega}_i, \vec{\omega}_o) L_i(\mathbf{x}, \vec{\omega}_i) \cos \theta_i d\vec{\omega}_i.$$

- (ii) Diffuse reflection.

$$L_o(\mathbf{x}, \vec{\omega}_o) = f_r(\mathbf{x}) \int_{H^2} L_i(\mathbf{x}, \vec{\omega}_i) \cos \theta_i d\vec{\omega}_i.$$

- (iii) Reflection is constant at all geometry points.

$$L_o(\mathbf{x}, \vec{\omega}_o) = f_r \int_{H^2} L_i(\mathbf{x}, \vec{\omega}_i) \cos \theta_i d\vec{\omega}_i.$$

- (iv) The incoming light is the same for all incident angles but there can be occluders.

$$L_o(\mathbf{x}, \vec{\omega}_o) = f_r L_i \int_{H^2} V(\mathbf{x}, \vec{\omega}_i) \cos \theta_i d\vec{\omega}_i,$$

where V is the visibility function.

- (v) There is a single object and we ignore self-occlusions.

$$L_o(\mathbf{x}, \vec{\omega}_o) = f_r L_i \int_{H^2} \cos \theta_i d\vec{\omega}_i.$$

The Rendering
Equation -
Inverse Rendering

- (b) Under certain assumptions, we can simplify the rendering equation to the following: $L_o(\mathbf{x}, \vec{\omega}_o) = f_r \int_{H^2} L_i(\vec{\omega}_i) \cos \theta_i d\vec{\omega}_i$. Assume we have a single known geometry for which we can compute surface normals $\mathbf{n}(\mathbf{x})$. Given three measurements from the surface $c_k = L_o(\mathbf{x}_k, \vec{\omega}_o)$ at known points $\mathbf{x}_k$, compute the reflected radiance $L_o(\mathbf{x}, \vec{\omega}_o)$ at an arbitrary point $\mathbf{x}$ on the surface. [Hint: If you encounter a linear system $\mathbf{Ax} = \mathbf{b}$, assume $\mathbf{A}^{-1}$ is known.] [4 marks]

Answer:

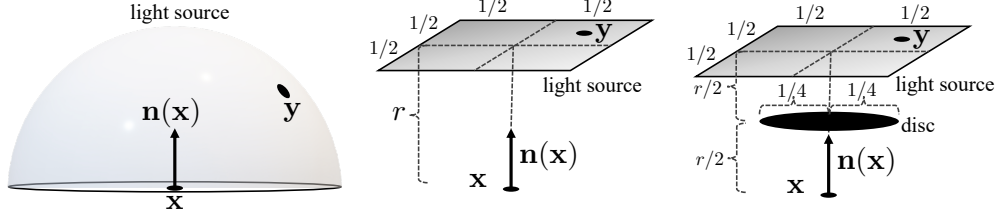
$$L_o(\mathbf{x}, \vec{\omega}_o) = f_r \int_{H^2} L_i(\vec{\omega}_i) \cos \theta_i d\vec{\omega}_i = \left(f_r \int_{H^2} L_i(\vec{\omega}_i) \vec{\omega}_i d\vec{\omega}_i \right)^T \mathbf{n}(\mathbf{x}) = \mathbf{l}^T \mathbf{n}(\mathbf{x})$$

$$\mathbf{l}^T \mathbf{n}(\mathbf{x}_k) = c_k$$

Stacking these, we get the linear system $\mathbf{N}\mathbf{l} = \mathbf{c}$, where the k^{th} row of $\mathbf{N}$ is $\mathbf{n}(\mathbf{x}_k)^T$. Hence, $\mathbf{l} = \mathbf{N}^{-1}\mathbf{c}$. Given a new point $\mathbf{x}$, we know $\mathbf{n}(\mathbf{x})$, and thus the reflected radiance is $\mathbf{l}^T \mathbf{n}(\mathbf{x})$.

The Rendering
Equation -
Surface Area
Form

- (c) Recall that for direct illumination, we have the following form for the reflection equation: $L_r(\mathbf{x}, \mathbf{z}) = \int_A f_r(\mathbf{x}, \mathbf{y}, \mathbf{z}) L_i(\mathbf{x}, \mathbf{y}) G(\mathbf{x}, \mathbf{y}) dA(\mathbf{y})$. Starting from this form, provide answers to the following in terms of the provided parameters.



- (i) Assuming a surface point $\mathbf{x}$ lit by a hemispherical light source with constant emitted light radiance L_i (left in the figure), derive the equation for the reflected light at $\mathbf{x}$ if the BRDF is a constant c . [Hint: $\int \cos \theta \sin \theta d\theta \equiv -0.5 \cos^2 \theta$] [4 marks]
- (ii) Assuming a surface point $\mathbf{x}$ with a constant BRDF c lit by a planar 1×1 square light source distance r away (middle of the figure) where the emitted light radiance decays as $L_i(\mathbf{x}, \mathbf{y}) = 1/\cos^4 \theta$ with θ the angle between the emitted light direction and plane normal, derive the equation for the reflected light at $\mathbf{x}$ (the surface normal $\mathbf{n}(\mathbf{x})$ is orthogonal to the planar light source). [4 marks]
- (iii) Now assume we put an occluding opaque disc of radius 0.25 parallel to the light plane and centred at $\mathbf{x}$ between the light plane and $\mathbf{x}$ at a distance $r/2$ from $\mathbf{x}$ (right in the figure). Derive the equation for the reflected light at $\mathbf{x}$. [3 marks]

Answer:

- (i) We can write:

$$L_i(\mathbf{x}, \mathbf{y}) = L_i,$$

$$G(\mathbf{x}, \mathbf{y}) = V(\mathbf{x}, \mathbf{y}) |\cos \theta_i| |\cos \theta_o| / \|\mathbf{x} - \mathbf{y}\|^2 = |\cos \theta_i| / r^2,$$

$$f_r(\mathbf{x}, \mathbf{y}, \mathbf{z}) = c.$$

Thus:

$$\begin{aligned} L_r(\mathbf{x}, \mathbf{z}) &= \int_A f_r(\mathbf{x}, \mathbf{y}, \mathbf{z}) L_i(\mathbf{x}, \mathbf{y}) G(\mathbf{x}, \mathbf{y}) dA(\mathbf{y}) \\ &= L_i c \int_A |\cos \theta_i| / r^2 dA(\mathbf{y}) \\ &= L_i c \int_H \cos \theta_i dA(\mathbf{y}), \\ &= L_i c 2\pi \int_0^{\pi/2} \cos \theta \sin \theta d\theta = L_i c \pi \end{aligned}$$

— *Solution notes* —

(ii) For the geometry term G , we have $\cos \theta_i = \cos \theta_o = \cos \theta$ and $\|\mathbf{x} - \mathbf{y}\| = r / \cos \theta$, thus

$$G(\mathbf{x}, \mathbf{y}) = V(\mathbf{x}, \mathbf{y}) |\cos \theta_i| |\cos \theta_o| / \|\mathbf{x} - \mathbf{y}\|^2 = V(\mathbf{x}, \mathbf{y}) \cos^2 \theta / (r / \cos \theta)^2 = V(\mathbf{x}, \mathbf{y}) \frac{\cos^4 \theta}{r^2}.$$

We further have $f_r(\mathbf{x}, \mathbf{y}, \mathbf{z}) = c$ and $L_i(\mathbf{x}, \mathbf{y}) = 1 / \cos^4 \theta$. Plugging these:

$$\begin{aligned} L_r(\mathbf{x}, \mathbf{z}) &= \int_A f_r(\mathbf{x}, \mathbf{y}, \mathbf{z}) L_i(\mathbf{x}, \mathbf{y}) G(\mathbf{x}, \mathbf{y}) dA(\mathbf{y}) \\ &= c \int_A (1 / \cos^4 \theta) V(\mathbf{x}, \mathbf{y}) (\cos^4 \theta / r^2) dA(\mathbf{y}) \\ &= c \int_A V(\mathbf{x}, \mathbf{y}) dA(\mathbf{y}) / r^2 \\ &= c / r^2. \end{aligned}$$

(iii) As derived above, $L_r(\mathbf{x}, \mathbf{z}) = c \int_A V(\mathbf{x}, \mathbf{y}) dA(\mathbf{y}) / r^2$. When projected onto the light plane, the disc of radius 0.25 becomes a disc of radius 0.5. Hence, we integrate over a unit square cut by a disc touching its edges, giving the area $1 - \pi 0.5^2$. The final result is thus $c(1 - \pi 0.5^2) / r^2$.
